

Zahra TehraniNasab

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in [zahra-tehraninasab](#)

Education

McGill University

M.Sc. Electrical Engineering (Thesis), GPA: 3.93/4

Montreal, Canada

Aug 2023 – Present

Sharif University of Technology

B.Sc. Computer Science, GPA: 17.64/20

Tehran, Iran

Sep 2018 – Feb 2023

National Organization for Development of Exceptional Talents (Sampad)

High School Diploma, GPA: 19.99/20

Shiraz, Iran

Sep 2015 – June 2018

Honors & Awards

Awarded the Mila EDI Scholarships - Women in AI Category, Valued at CAD\$10,000, MILA (Quebec AI Institute), 2024-2025

Awarded the Graduate Excellence Fellowship (GEF), Valued at CAD\$4,500, McGill University, 2023
Third Prize, MICCAI Educational Challenge (2025), MICCAI Educational Challenge for a blog post on generative models in medical imaging.

Ranked 229th Among More Than 140,000 (Top 0.17%) Participants, Iranian Nationwide University Entrance Exam, Mathematics and Physics Discipline, 2018

Publications

Pixel Perfect MegaMed: A Megapixel-Scale Vision-Language Foundation Model for Generating High Resolution Medical Images, MICCAI (2025) DGM4MICCAI Workshop Proceedings, Zahra TehraniNasab, Hujun Ni, Amar Kumar, Tal Arbel, [Project Website](#)

Language-Guided Trajectory Traversal in Disentangled Stable Diffusion Latent Space for Factorized Medical Image Generation, CVPR (2025) MIV Workshop Proceedings, Zahra TehraniNasab*, Amar Kumar*, Tal Arbel, [Project Website](#)

Pixels Under Pressure: Exploring Fine-Tuning Paradigms for Foundation Models in High-Resolution Medical Imaging, MICCAI (2025) ELAMI Workshop Proceedings, Zahra TehraniNasab, Amar Kumar, Tal Arbel, [Arxiv Link](#)

PIKACHU: Prototypical In-context Knowledge Adaptation for Clinical Heterogeneous Usage, Under Review, Amar Kumar, Zahra TehraniNasab, Emily Kaczmarek, Tal Arbel

RL4Med-DDPO: Reinforcement Learning for Controlled Guidance Towards Diverse Medical Image Generation using Vision-Language Foundation Models, MICCAI (2025) Main Track, Parham Saremi*, Amar Kumar*, Mohammed Mohammed, Zahra TehraniNasab, Tal Arbel, [Project Website](#)

Discovering Latent Knowledge Graphs for Capturing Diversity in Conditional Image Generation, NeurIPS (2025) Main Track, Bailey Trang, Parham Saremi, Alan Q. Wang, Fangrui Huang, Zahra TehraniNasab, Amar Kumar, Tal Arbel, Li Fei-Fei, Ehsan Adeli

Towards Reliable Human Pose Forecasting with Uncertainty, IEEE Robotics and Automation Letters, S. Saadatnejad, Z. TehraniNasab*, M. Mirmohammadi*, M. Daghyani*, P. Saremi*, Y. Zoroofchi Benisi*, A. Alimohammadi*, T. Mordan, A. Alahi, [Arxiv Link](#)

Research Experience

Intern, Mila - Quebec AI Institute

Multimodal Learning

Montreal, Canada

Present

- Developing a multimodal prospectivity mapping framework for geospatial mineral exploration.

M.Sc. (Thesis) Student, McGill University

Generative Modeling, Explainability, Fairness, Synthetic Data Generation
Supervisor: Prof. Tal Arbel

Montreal, Canada

Sep 2023 – Present

- Developing fine-tuning methods for Vision Language Foundation Models for generalizability in healthcare.
- Understanding the properties of latent subspaces and controlled traversal in them using language guidance.

Research Intern, École polytechnique fédérale de Lausanne (EPFL)

Human Pose Prediction, VITA Lab

Supervisors: Dr. Alexandre Alahi and Saeed Saadatnejad

Remote Internship

Dec 2021 – Jul 2022

- **Improved state-of-the-art models for about 2% to 5%** by integrating Predictive Uncertainty Awareness.
- Collaborated with five colleagues on developing and maintaining a general **deep learning library** for Human Pose Prediction using Python and PyTorch.

Research Intern, University of California Irvine (UCI)

Food Computing

Supervisors: Prof. Ramesh Jain and Prof. Hamid R. Rabiee

Remote Internship

Dec 2021 – Jul 2022

- Joint research project between the the **University of California Irvine (UCI) and the Sharif University of Technology (SUT)**
- Developed and experimented with a hierarchical visual feature extractor to predict ingredients and categories of food from an image.
- Developed and experimented with a Graph-LSTM network to predict food categories from the food recipe graph.

Related Courses

CIFAR Deep Learning + Reinforcement Learning (DLRL) Summer School (2025)

[Certificate](#)

Statistical Computer Vision, McGill University

Grade: 4/4

Natural Language Processing, McGill University

Grade: 4/4

Machine Learning In Network Biology, McGill University

Grade: 4/4

Representation Learning, University of Montreal

B.Sc. Thesis on Virtual Try-on, Sharif University of Technology

Grade: 20/20

Supervisors: Dr. Mohammad Hadi Foroughmand and Dr. Mostafa Kamali Tabrizi

- Investigated and wrote a review on the methods and ideas proposed in the recent virtual try-on papers.
- Studied and wrote a review on the following concepts:
 - Conditional Generative Adversarial Networks (**C-GANs**) — **Knowledge Distillation** (Teacher-Student Networks) — Human Pose Estimation — Inpainting — Optical Flow

Deep Learning, Neuromatch Academy — International Summer School

[Certificate](#)

Computer Vision, Sharif University of Technology

Grade: 19.3/20

Natural Language Processing, Sharif University of Technology

Grade: 20/20

Image Processing, Sharif University of Technology

Grade: 19.7/20

Machine Learning, Sharif University of Technology

Grade: 18.8/20

Neural Networks and Deep Learning, Coursera — Online Course

[Certificate](#)

Teaching Assistant

Applied Machine Learning, McGill University

Jan 2025 – April 2025

Fundamentals of Machine Learning, McGill University

Aug 2024 – Dec 2024

Artificial Intelligence, Sharif University of Technology

Feb 2021 – Jul 2022, Sep 2022 – Dec 2022

Computer Systems, Sharif University of Technology

Sep 2021 - Dec 2021

Networking, Sharif University of Technology

Feb 2021 – Jul 2021, Feb 2022 – Jul 2022

Skills

Programming languages: Python | Typescript | F# | Java | Matlab | Octave

Operating systems and developer tools: Linux | Windows | Git | GitHub | Tmux

Deep Learning skills and tools: Vision-Language Foundation Models (VLMs), Large Language Models (LLMs), Generative Modeling, Computer Vision, Natural Language Processing, Sequence Modeling, Working with GPU clusters, Working with distributed processing on multiple GPUs and Devices, Pytorch, Numpy, Pandas, Sklearn, Matplotlib, HuggingFace, CometML